CS5003 –Data Structures and Specialist Programming Coursework (Group Programming Assignment + Report)

This coursework carries 40% of the module's overall mark. Students are expected to work in a group (3 students) to complete the work described in this document.

Please note the rules on plagiarism and using Generative AI tools

Any material which is a direct copy from someone else (student, GenAI or other source) or a close paraphrase/code must be indicated where it is quoted. You must indicate where quotations or close paraphrases occur, e.g., by using italics or quotation marks. It is not sufficient to show the source in a list of references/bibliography.
You should avoid copying and pasting any AI-generated output directly into your work. You could use Gen-AI tools for learning purposes but not to do the job instead of you (i.e., your code should be written by you, and you should fully understand what each line of code does).

If you are unclear, please discuss your examples with your workshop tutor or the module leader. **Plagiarism is a serious offence, and conviction for plagiarism may lead to serious action or suspension from the University. Please see the section on Academic Misconduct on the university website to learn more, in addition to the following link about AI usage: Link**

Software Development Task

Your task is to create a **Java web application agreed with your module tutor, using the techniques and guidelines you have been given in the module. Your website has to contain dynamic pages and at least one static page (e.g. contact us page**. **You cannot get a pass mark if all pages are static pages.**

The web application must use Java client-server technology and a backend database. It must demonstrate the features that you have been taught in the course, in particular **Java beans, JavaServer™ Faces, Ajax, and accessing a database in a web application.** The database could be any of Java DB, Apache Derby, MS Access, MySQL, Oracle, and MS SQL. Your database must be completely different to any sample database supplied during the module, by Deitel.com or netbeans.org websites – in terms of table and field names. This rule does not apply to login tables, if any. You will need to discuss the tools, website map and database tables with your tutor and get agreement.

The deliverables

This coursework requires the submission of the following group work,

1- **The code:** Upload your complete artefact/code (including the database SQL Script, and database zipped folder) to a GitHub repository https://github.com/ and place the repository link in your final report (in the cover page). **Make sure your code is accessible (sign out of GitHub and test the link).**

As a backup, please upload your software artefact to WebLearn. To do this, you will need to create a new folder on your computer, in which you can place the Java web project folder, the database SQL Script file, and the database zipped file. Zip/Compress the whole folder and submit it. **Follow this naming format: 'YourStudentID_StudentLastName_ Artefact.zip'.**

2- **The report** in MS-Word compatible or PDF format. **Follow the naming format: 'YourStudentID_StudentLastName_Report.pdf'**

3- **The Video Demo**: A recorded software demo in mp4 format (5 minutes max), using Panopto application. The demo should demonstrate the testing of your coursework implemented functionalities (with some focus on the criteria listed in the TESTING AND IMPLEMENTATION table below). **Follow this naming format: 'YourStudentID_StudentLastName_Demo.mp4'.**

Report Details and Marking Scheme (up to maximum 1500 words, excluding the cover page, references, and the code or testing printouts)

Please be concise in your report (clear statements in fewer words), use bullets and numbering for text descriptions instead of long paragraphs. Use tables and diagrams where applicable.

Your report should include the following sections (section names highlighted):

- COVER PAGE, contains
  - The title of your project (i.e. the selected topic)
  - The group member names and IDs
  - The link for your code repository on GitHub
  - Steps to run your project. Example: download the project folder/zipped file, open the project in Netbeans, so on.

    You will need to give some guidelines on how the database can be integrated and any specific tool versions require consideration to run your software (e.g., which server is used?). **Note that these instructions are not a 'User Manual' of your application. Concisely list the steps in bullet points or a flowchart (not in a paragraph)**

- INTRODUCTION

A brief description of your application, including the purpose, audience, key features and the implemented functions.

- SITE MAP

Site map diagram showing the whole structure of your site – how the pages link together.

- DATABASE

  - Place the table relationship diagram (ERD) that describes the tables, their fields and the relationship between these tables. Note that your database must have at least one table with at least 3 field types.
  - Place a screenshot of the tables you created, displaying some records populated with a sample of meaningful test data.

- TESTING AND IMPLEMENTATION:

Place screenshots (live, maximum of 2 screenshots per A4 side with **annotation**) **to show a clear example of test data and test results** for each of the required system functionalities as below,

Tested Functionalities and Implementation	Expected Screenshot/(s) (i.e. evidence)
Menu structure/navigation	Screenshot (s) of a maximum of 2 pages that show your home page and the implemented Menu structure or navigation tools to other pages.
CRUD operations	For each of the 4 CRUD operations, - place screenshot(s) to show the operation, including any testing data required through your website pages; - followed by screenshot (s) to show the change in the corresponding table records in your database.
Validation	- A Screenshot to validate at least one field on the website pages. - A Screenshot of the corresponding code implementation from your project files.
Ajax e.g. on a search screen/form	- A Screenshot of testing Ajax on your website pages. - A Screenshot of the corresponding code implementation from your project files.
Session Tracking	- Screenshot(s) of testing Session tracking. For example, the user name appears

	across the pages.
UI components	- Screenshot of an example code implementation of a page with UI components

- WORK MANAGEMENT and REFLECTION

  - Your reflection should clearly describe how the work was distributed among team members (**including each member's name, their specific responsibilities, and the nature of their contribution**). Comment on whether the team was satisfied with everyone's contribution. Please note that **all group members should actively contribute to the software implementation (coding), and best efforts should be made to reflect this contribution in the submitted artefact (e.g., added comments)/GitHub activity (e.g., commits).**

  - Your reflection should cover your experience during the coursework, including lessons learned, challenges you faced and how you overcame them, and insights you can take forward. For example, if you were to redo the project, what changes would you make based on your learning, and how the project has contributed to your Continuous Professional Development (CPD).

- ARTEFACT DESCTIPTION

  Use a table format to list the project file names, and their corresponding concise description, including Java classes, pages, a non-trivial CSS script, and any other files which are part of your submitted work. You must explain the purpose of, and the technologies used.

- REFERENCES

Important notes:

1. Failure to follow these instructions e.g. not using Java, client-server web application, or databases, using pages from multiple websites, or using ready-made solutions, will put you at risk of failing this assessment.
2. Before the submission, please check that your application runs in at least 2 PCs!
3. Make sure all text and diagrams are visible within the pages' margins before you submit your report file.

MARKS WILL BE DEDUCTED IF THE CODE IS NOT SUBMITTED OR IS NOT ACCESSIBLE.

Marking Scheme

Cover page	5%
Introduction and Site Mar (Content, functionality and usefulness of the application)	8%
Database tables and sample data	5%
Testing and implementation	40%
Work management and reflection	10%
Artefact, description and structure (e.g. Pages, JavaBeans, MVC, etc.)	10%
Report clarity, format, and references	5%
Adherence to the coursework requirements (e.g., using the appropriate tools to implement the coursework)	10%
User interface design	7%
Total	100%

Get Started.. Don't leave it to the last minute. I am sure you can do a great job.
"A little progress every day can adds up to big results"
With my best wishes in your project!

Dr. Mona Abdelgayed

Appendix: Guidelines for implementing this coursework

1- Choose a topic of interest from the indicative list released with the coursework. If you wish to work on another topic, you will need to seek approval from your tutor before you proceed on.

2- Register your topic on Weblearn following the module leader's instructions.

3- Prepare a written plan including: Website map including dynamic and static pages, database tables, data you will include in both pages and database, the website functionalities and features you plan to implement to fulfil the assignment requirements.

4- Discuss your plan with your tutor and seek feedback.

   Note: Your website has to contain at least 2 pages. The number of pages your website should include is based on the implemented features. Choose a number that fulfils the implementation of the required features (examples: CRUD operations, Ajax, Menu/navigation, etc; check the 'Testing and Implementation' table and marking scheme for the required features) based on your application.

5- Design, create your application's database and populate it with sample data for testing. Your database must have at least a table with at least 3 field types.

6- Apply the lessons you learned in the module about the three-tier and MVC architectures to create the website, including: pages and CSS files, business logic and a connected database.

7- Make sure you apply the requirements according to the marking scheme.

8- Before Submission, check that your application runs in at least 2 PCs!

LONDON METROPOLITAN UNIVERSITY

School of Computing and Digital Media

Marking Criteria for Coursework

Marking criteria	Mark range
An exceptional submission, with extensive and detailed knowledge based on a high level of additional background research; a high degree of critical analysis, evaluation and original insight; excellent organization and presentation.	90 – 100
In addition to the requirements for grades of 70-79% (below), an outstanding submission incorporating a high level of originality, depth and critical insight and going well beyond expected work.	80 – 89
An excellent submission, going beyond expected work; commanding understanding and appreciation of the central points; well-written and effectively structured; evidence-based, critical and logical analysis; comprehensive and correct referencing of sources.	70 – 79
A very good and comprehensive submission that fulfills the assignment brief; relevance and accuracy; clear structure and evidence-based; a sound grasp of the subject and ability to think about it effectively and critically; correct referencing of sources.	60 – 69
A solid submission that fulfills most of the assignment brief; adequate structure; mostly accurate, with few errors or omissions; some limitations in scope, critical thinking and argument; a consistent attempt at referencing sources.	50 – 59
A basic but incomplete submission, with limited relevant information; lacks logical and coherent structure, with some significant errors or omissions; contains sparse and/or irrelevant information and lacks an evidence-based approach; some limitations with referencing of sources; poor grammar and spelling.	40 – 49
An unsatisfactory submission, with some substantial errors, omissions or irrelevancies; barely acceptable amount of information relevant to the question; poor structure, presentation and expression and referencing of sources.	30 – 39
An unsatisfactory submission, that doesn't meet the learning outcomes with many substantial errors, misconceptions, omissions or irrelevancies; little information relevant to the question; very poor structure, presentation and expression.	1 – 29
Non-submission or submission of work which cannot be given any credit (e.g., blank submission, incorrect assignment).	0